FINAL CASE

Spare Part Intelligence & Inventory Optimization

AI-Powered Forecasting & Dynamic Optimization



Spare Part Intelligence & Inventory Optimization

AI-Powered Forecasting & Dynamic Optimization

WHAT'S HAPPENING

1 THE TRIGGER

The Right Items aren't Available at the Right Time

When a shutdown happens, the **critical part we need is out of stock**, while slow-moving items we don't need sit available on the shelf.

Result:
rush procurement, higher cost, downtime risk.

2 THE CAUSE

Planning lags behind

Stock is planned reactively: **failures, maintenance, shutdowns and lead times**, not around what really drives demand

3 THE DILEMMA

Cost vs Risk

Too much stock ties up capital and raises carrying costs.

Too little stock means stockouts, emergency buying and production loss.

OBJECTIVE

Establish an AI-powered spare parts optimization solution to improve forecasting, enable dynamic inventory decisions, reduce excess inventory and working capital, and maintain critical spare-part availability.



Spare Part Intelligence & Inventory Optimization

AI-Powered Forecasting & Dynamic Optimization

THE PROBLEM

The Company lacks an **integrated and predictable approach** to forecasting spare part demand and inventory level setting

WHY IT HAPPENS

1 Static planning

- Relies on historical consumption
- Static Min–Max parameters

2 Suboptimal inventory balancing

- Inventory decisions **do not adequately balance** part availability, carrying cost, and downtime exposure
- Criticality and replenishment lead time are **not consistently** considered into stocking decisions

RESULTING IMPACT

- **Emergency procurement**
- **Higher purchasing & carrying costs**
- **Excess tied-up working capital (High inventory investment)**
- **Prolonged downtime & production losses**



Spare Part Intelligence & Inventory Optimization

AI-Powered Forecasting & Dynamic Optimization

EXPECTED SOLUTION

1 Demand Forecasting

Predict future spare part requirements based on historical consumption, equipment failures, maintenance activities, and operating conditions.

2 Dynamic Inventory Optimization

Recommend optimal Min–Max levels, Safety Stock, Reorder Points, and replenishment quantities based on changing demand and lead times.

3 Inventory Health Monitoring

Detect excess, slow-moving, non-moving, and potentially obsolete spare parts.

4 AI Recommendation & Cost–Risk Optimization

Recommend which parts should be reordered, maintained, reduced, or monitored by balancing inventory cost against stockout, downtime, and production-loss risk.

EXPECTED AI CAPABILITIES

The solution should move beyond forecasting by providing actionable recommendations such as **Stockout Risk, Overstock Risk, Reorder Recommendation, Slow-Moving Alert, Dynamic Inventory Parameters, and Cost–Risk Optimization Scenarios.**



Final Case

KEY QUESTIONS

Participants must address the following two key questions in the deck, either within the Proposed Solution or Business Impact section.

1 **How can the Company improve demand predictability for spare parts with irregular and uncertain consumption patterns?**

MAIN FOCUS

Build **AI-powered spare part demand forecasting** and demand intelligence to anticipate future requirements and improve spare part availability.

PARTICIPANTS SHOULD CONSIDER

- Historical spare part consumption and usage patterns
- Equipment failure frequency and failure history
- Planned maintenance activities
- Preventive maintenance schedules
- Shutdown and turnaround activities
- Installed equipment population
- Seasonal, production, and operating conditions
- Abnormal or unexpected consumption patterns
- Procurement lead time and supply variability



Final Case

2 **How can the Company dynamically optimize spare part inventory while balancing inventory cost, availability, and operational risk?**

MAIN FOCUS

Develop **AI-enabled inventory optimization and decision recommendations** that determine appropriate stock levels and actions based on future demand, inventory conditions, and operational consequences.

PARTICIPANTS SHOULD CONSIDER

- Which critical spare parts are at risk of stockout
- Which spare parts are potentially overstocked
- Which items should be reordered and when
- Recommended reorder quantities and dynamic Min–Max levels
- Which items are becoming slow-moving or dead stock
- Inventory carrying and holding costs
- Procurement cost and replenishment lead time
- Equipment and spare part criticality
- Potential downtime and production losses caused by spare part unavailability
- Trade-off between reducing inventory cost and increasing stockout or downtime risk
- Scenario simulation of increasing or reducing inventory levels



Final Case

BASELINE DATA

You can retrieve the baseline data from this link:
https://bit.ly/CALIBER2026-DataFinalCase

Besides the provided baseline data, participants may propose additional data sources where relevant. However, every additional data source should be supported by clear business justification. **A well-justified and high-quality data architecture is preferred over simply integrating more data.**

1 Spare Part Master Data

File: BoM (Bill of Material)

Defines spare part characteristics and existing inventory control parameters.

2 Spare Part Consumption History

File: Movement Material

Identifies historical, intermittent, and abnormal demand patterns for forecasting.

3 Current Inventory Position

File: BoM (Bill of Material)

Shows current stock availability, item min-max, and potential shortage or excess conditions.

4 Equipment Criticality & Failure History

File: Notif History; FuncLoc List (Functional Location List)

Connects spare part requirements with equipment reliability, failure probability, downtime exposure, and operational risk.

5 Procurement & Lead-Time Data

File: PR PO

Supports replenishment and inventory optimization using lead time, cost, and supply reliability.

6 Maintenance Plan

File: Mitem & Mplan

Provides visibility of known future spare part demand from planned, preventive, overhaul, and shutdown activities.



Submission Format

Please submit a clear representation of your proposed solution to demonstrate the concept, user flow, and expected output. To be eligible for final presentation, you must submit these 2 materials:

1 Prototype

- Minimum viable prototype (MVP, not static HTML) published to Vercel (https://vercel.com)
- Database using SQL Lite
- Complete source code and database in .zip file
- Screenshot (.png/.jpg/.jpeg) timestamp of the latest deployed version in Vercel

2 Presentation deck

- See next slide for the format details

SUBMISSION DEADLINE

Thursday, 5 November 2026 at 06.00 AM

Submission link will be sent to participants later on CALIBER 2026 Whatsapp Group and your email.

All materials should be delivered in English.

Proposal must strictly follow the guidelines provided. Non-compliance will directly affect the evaluation outcome.



Presentation Deck Format

Slide	Required Content
1	**Cover** Identify your team's name and team members' names
2	**Background & Problem Statement** Explain the current challenges, supporting facts, and why the problem is important to solve.
3	**Solution** Present the proposed solutions to solve the problem and key technologies/features that make your solutions unique.
4	**Business Impact** • Success metrics: how success will be measured • Expected benefits to the business
5	**Roadmap & Risk Management** • High-level timeline showing key activities, outputs/deliverables, and stakeholders involved in each phase. • State the key risks and how to mitigate them.
7	**Team Profile & Key Technologies** • Provide role and contribution of each team member through below table.

No.	Name	Major	Semester	Area of expertise	Contribution
1	[Example-1]	Computer Science	5		Formulate the idea, etc.
2	[Example-2]	Chemical Engineering	5		Elaborate background and problem statement, etc.

	• Mention the **AI models** and tools used to build the proposed solutions.
	Appendix (optional) Supporting references or additional information

Good Luck!

